



SOP #18

Protein Extraction

Note: Protein Extraction can be done using an automated “bead-beater” machine which uses commercially available, pre-filled tubes with ceramic, mineral stone or metal beads in it. Ours is the “FastPrep” by MP Biomedicals - there are 2 sizes of tubes: 2 mL (\$3/ea) and 15 mL (\$18/ea). Alternatively, a sonication method is available if the bead-beater is unavailable or too expensive, although this alternative method is not recommended due to variation in tube-to-tube handling.

References for extractions:

Diatoms: PLoS ONE 8(1): e55562. doi:10.1371/journal.pone.0055562

Picocyanobacteria: (Life 2014, 4, 1-x manuscripts; doi:10.3390/life40x000x)

Prasinophytes: Photosynth Res DOI 10.1007/s11120-016-0310-6

Extraction Protocol for FastPrep:

1. Prepare 1 mL (per pellet) of 1x protein extraction buffer (**1x PEB**). Multiply all of these following values by the number of pellets you have to get the actual volumes to be used. Note: this calculation will give you excess buffer, which you must use the day it is made. Pefabloc is a protease inhibitor, and it loses reactivity over 24 hrs.
1X PEB (1 mL):
250 μ L 4X PSB (see recipe below)
20 μ L 50X AEBSF (aka “Pefabloc”)
730 μ L ddH₂O
2. Onto each frozen pellet pipette between 400 μ L and 750 μ L PEB (500 μ L is a good starting point, and can be increased or decreased depending on the total protein assay outcome). Once the pellets have begun to thaw, homogenize them gently using a pipettor with a cut-off pipette tip. Once homogenized, store them fully immersed in the icebox.
3. Pipette the solution into a labelled 2 mL microtube with the beads (Sigma Z763799 or MP Biomedicals 116913050). Alternative method for glass fiber filters (GFF by Millipore Cat. No. APFB02500): add the filter and extraction buffer directly to the FastPrep lysing tube with tweezers. Work quickly. Label tubes on top and on the sides.
4. Lyse for 60 seconds at 6.5m/s in FastPrep-24.
5. Cool microtubes on ice for 60 seconds.
6. Repeat steps 4 and 5 twice (the number of FastPrep rounds may have to be optimized).
7. Centrifuge microtubes in a benchtop centrifuge at 14,800g for 5 minutes.
8. Pipette each supernatant into clean 0.6 mL microtube, label.
9. Homogenize with gentle pipetting, and divide each supernatant into 125 μ L aliquots or less. (1 for total protein content, one for immunoquantitation, the rest in case multiple immunoquantitations are necessary). Label the 4+ microtubes with 1/4 to 4/4 (in addition to date, initials and content). Volumes do not have to be exactly divided, as the volume recovered during the spin step varies. Freeze extracts in -80°C freezer.

Note: Each time an aliquot of extract is thawed for use, document it. The tube used for total protein content (1/4) can be freeze-thawed as many times as necessary, as the assay does not require protein integrity. Any protein quantitations (Western blotting or Wes) should be done on a new, never-thawed aliquot, as multiple freeze-thaw cycles will degrade proteins differentially, skewing quantitation results.

4X Protein Solubilization Buffer (PSB) >Low Tris Recipe (2015)

0.136 g TRIS Base (powder)

0.133 g TRIS HCl (powder)

0.8 g LDS (powder - harmful if inhaled, measure out while in the hood)

2.5 ml of ddH₂O to dissolve

4 g glycerol - molecular biology grade to avoid interference with protein assays

40 μ L 0.5M EDTA (lab stock solution in freezer)

Fill to 10 ml with ddH₂O. This should be enough for ~80 samples, and is stable at room temp.